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Vehicle software change control system

Abstract

A vehicle software release management system enables receipt of a vehicle software package and an associated test plan to generate a versioned software artifact set. The vehicle software release management system may initiate a workflow to manage testing, certification, and deployment of the software package based on the versioned software artifact set. The vehicle software release management system may facilitate testing of the vehicle software package, including generating test scripts and configuring the testing environment according to test configurations. The vehicle software release management system may facilitate certification of the vehicle software package according to the versioned software artifact set and manage the deployment/rollback to software package destinations.

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Background/Summary

BACKGROUND

(1) Various vehicle systems are equipped with software that allow components of the vehicle to function. For example, vehicle software may be used to control vehicle control systems, vehicle infotainment systems, vehicle environmental control systems, navigation systems, etc. For various reasons, it may be advantageous for vehicle manufactures to update vehicle software, such as to correct software issues, improve performance, add new features, protect against discovered vulnerabilities, prevent unwanted access to the vehicle control systems, amongst various other uses.

Over the course of the vehicle's lifetime, there may be various reasons to update vehicle software for various ones the vehicle components and such updates may require proper versioning and vetting before being deployed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 illustrates a vehicle software release management system that receives a vehicle software package and a test plan that are used to generate a versioned artifact set. The vehicle software release management system allows performance of testing and certification of the software package according to the software artifact set and distributes the certified vehicle software package to one or more destinations, according to some embodiments.

(2) FIG. 2 illustrates a more detailed view of a vehicle software release management system that receives a vehicle software package and a test plan that are used to generate a versioned artifact set for a vehicle model that may be stored in a versioned artifact set storage, according to some embodiments.

(3) FIG. 3 illustrates a more detailed view of a vehicle software release management system that initiates a workflow to perform testing of a vehicle software package according to the associated software artifact set for a vehicle model and incorporates the vehicle software test metadata (e.g., testing parameters and testing results) into the software artifact set for the vehicle model, according to some embodiments.

(4) FIG. 4 illustrates a more detailed view of a vehicle software release management system that initiates a workflow to perform certification of a vehicle software package corresponding to an associated software artifact set for a vehicle model. The release management system also incorporates the vehicle software certification metadata into a software artifact set that is stored by the vehicle software release management system for the vehicle model, according to some embodiments.

(5) FIG. 5 illustrates a more detailed view of a vehicle software release management system that initiates a workflow to deploy a vehicle software package selected from a versioned artifact set storage to a software package destination, according to some embodiments.

(6) FIG. 6 illustrates a more detailed view of a vehicle software release management system that receives a request to rollback a vehicle software package and initiates a workflow to transmit a software package included in a software artifact set that was previously deployed or roll back software resources from a software package destination for a previously deployed software package, according to some embodiments.

(7) FIG. 7 illustrates a flowchart of operations performed by a vehicle software release management system to receive a vehicle software package and a test plan, generate a versioned artifact set for the vehicle software package, initiate performance of testing, and send a request to certify the software package according to certification instructions included in the software artifact set, according to some embodiments.

(8) FIG. 8 illustrates a flowchart of operations performed by a vehicle software release management system to initiate deployment of a certified vehicle software package to a vehicle software package destination, according to some embodiments.

(9) FIG. 9 illustrates a flowchart of operations performed by a vehicle software release management system to initiate rollback of a deployed vehicle software package from a vehicle software package destination, according to some embodiments.

(10) FIG. 10 illustrates a block diagram illustrating an example computer system that implements some or all of the techniques described herein, according to some embodiments.

(11) While embodiments are described herein by way of example for several embodiments and

illustrative drawings, those skilled in the art will recognize that embodiments are not limited to the embodiments or drawings described. It should be understood that the drawings and detailed description thereto are not intended to limit embodiments to the particular form disclosed, but on the contrary, the intention is to cover all modifications, equivalents and alternatives falling within the spirit and scope as defined by the appended claims. The headings used herein are for organizational purposes only and are not meant to be used to limit the scope of the description or the claims. As used throughout this application, the word “may” is used in a permissive sense (i.e., meaning having the potential to), rather than the mandatory sense (i.e., meaning must). Similarly, the words “include,” “including,” and “includes” mean including, but not limited to.

DETAILED DESCRIPTION OF EMBODIMENTS

(12) The systems and methods described herein include techniques for implementing a vehicle software release management system that receives a vehicle software package, an associated test plan, and generates a versioned artifact set. The vehicle software release management system may furthermore oversee testing, certification, and deployment of the software package based on the versioned artifact set.

(13) For example, various vehicle systems may be embedded with software that enable vehicle systems and the components therein to function. Vehicle software may be required in controlling any number of vehicle functions ranging from controlling the brakes to receiving signals from a sensor. Updates to the vehicle software may be made periodically to maintain correct functioning of the software. Additionally, regular updates to the software may be necessary to fix bugs in the software, improve performance of the component, add new features, and/or protect against discovered security vulnerabilities. Management of the multiple software packages for the various components of a vehicle system poses a non-trivial challenge as the various packages need to be organized, tested, and certified before being deployed to a correct destination. Moreover, with regard to a specific software package, its testing environment, testing procedure, and certification procedure may be unique to its version of the software and/or to an associated vehicle model that the software package is to be deployed to. For example, in order for the software package to be deployed, a vehicle system owner, such as an original equipment manufacturer (OEM) may require the software package to be tested, and such testing may be required to be carried out in the correct operating system of the vehicle model having the correct test resources and dependencies that are associated to that unique version of the software package.

(14) A vehicle software release management system may provide a mechanism for vehicle software packages to be tested, certified, and then sent to a vehicle software release management system. The vehicle software release management system may also organize a given received vehicle software package into a versioned artifact set containing various software artifacts necessary to complete the vetting process of the received vehicle software package and enable deployment of the received vehicle software package (subsequent to certification) to a software package destination. In some embodiments, the vehicle software release management system may receive a test plan that describes a testing environment and processes that the software package is to undergo to perform the testing. Based on the received vehicle software package and the received test plan, the vehicle software release management system may generate a software artifact set. The generated software artifact set may comprise software artifacts or files that are associated with the particular version of the software package required in the testing and certification process, including configuration files, resource files, and/or test scripts, along with the original software package. The vehicle software release management system may generate one or more components of the software artifact set that are assembled. Once the software artifact set has been generated, the vehicle software release management system may store the artifact set in a versioned artifact set storage. In some embodiments, the vehicle software release management system may initiate one or more workflows to test and certify the software package.

(15) In some embodiments, the vehicle software release management system may enable testing of

vehicle software for a vehicle model as received with the vehicle software package by the management system or otherwise determined or indicated. This may be done as part of the testing workflow. The vehicle software release management system may generate various required software components of the vehicle software testing, such as test scripts, electronic control unit configurations, and/or dependencies required to execute the vehicle software test. The vehicle software package testing may be performed in a testing environment that may be a part of the vehicle software release management system, according to some embodiments. In other embodiments, the testing environment may be outside of the vehicle software release management system. Once the test has been performed according to the test plan, the vehicle software release management system may store the resulting testing metadata in the software artifact set. In some embodiments, the vehicle software release management system may enable certification of vehicle software for the vehicle model received by the management system as part of the certification workflow. The vehicle software release management system may send a request to a certification destination according to a format required by the certification destination. The certification destination may initiate its own separate internal certification workflow to determine whether the software package should be certified and allowed to be deployed. The resulting certification response is received by the certification system and further added to the software artifact set.

(16) In some embodiments, the vehicle software release management system may enable deployment and rollback of vehicle software for the vehicle model received by the management system. The vehicle software release management system may, in some embodiments, receive a request to deploy a certified vehicle software package to a software package destination. The vehicle software release management system may allow selection of the certified vehicle software package based on information provided by the software artifact set, including its testing metadata and its certification metadata. The vehicle software distribution system may format and transmit the requested certified vehicle software package for deployment to the deployment destination. In some embodiments, vehicle software release management system may furthermore receive a request to roll back a software package deployed to the software package destination. The vehicle software release management system may transmit a replacement certified vehicle software package determined by the vehicle software release management system, or, in some embodiments, transmit instructions to remove the portions attributed to the software package requested to be rolled back.

(17) FIG. 1 illustrates a vehicle software release management system that receives a vehicle software package and a test plan to generate a versioned artifact set. The vehicle software release management system allows performance of testing and certification of the software package according to the software artifact set and distributes the certified vehicle software package to one or more destinations, according to some embodiments.

(18) A vehicle software release management system **100** comprising a release management interface **110**, a versioned artifact set generation module **112**, a simulation configuration interface **120**, a versioned artifact set storage **130**, testing system **140**, certification system **150**, and a vehicle software distribution system **160** is illustrated in FIG. 1. In some embodiments, the vehicle software release management system **100** may receive a vehicle software package **111** through the release management interface **110**. The vehicle software package may be a piece of software used in the operation of one or more components in a vehicle as well as components in data pipeline of vehicles. For example, the software package may be related to software used in a vehicle data lake or in a vehicle data pipeline. In some embodiments the vehicle software package may be associated with a certain vehicle model or a given component model as further discussed in FIG. 2.

(19) In some embodiments, the vehicle software release management system **100** may receive a test plan **114** associated with the received vehicle software package via the simulation configuration interface **120**. The test plan may detail one or more requirements and parameters for testing the received vehicle software package. In some embodiments, a test plan may be a detailed document

that catalogs the test strategy, objectives, and/or the resources required in testing the vehicle software package as part of a vetting process. In some embodiments, the testing procedures described in various test plans may be different based on the vehicle model indicated for the software package, as further discussed in FIG. 2. In some embodiments, the simulation configuration interface **120** may configure the testing system **140** and/or a testing environment to conform to the testing environment described in the test plan. Once the vehicle software package and test plan have been received, the versioned artifact set generation module **112** generates a software artifact set **125**. The software artifact set may comprise software artifacts or files that are associated with a particular version of the software package required in the testing and certification process, including configuration files, resource files, and/or test scripts. Various components of the software artifact set may be generated by the release management interface **110**, versioned artifact set generation module **112**, and/or the simulation configuration interface **120**. For example, a test script associated with the received test plan for the software package may be generated by the simulation configuration interface.

(20) Once the software artifact set has been generated, the vehicle software release management system may store the artifact set in a versioned artifact set storage **130**. The versioned artifact set storage **130** may contain multiple software artifact sets for each of a plurality of received vehicle software packages. For example, the versioned artifact set storage **130** may contain software artifact set A **132**, software artifact set B **134**, and software artifact set N **136** each with an artifact set that contains a different vehicle software package. In some embodiments, the software artifact sets may differ not based on the vehicle software package, but based on other artifacts such as the test plan as further discussed in FIG. 5. For example, a same software package may reside in two different artifact sets, wherein the given software package was tested according to two different test plans. The vehicle software release management system **100** may initiate workflows to test and certify the received vehicle software package using the associated software artifact set.

(21) In some embodiments, the testing system **140** of the vehicle software release management system **100** may obtain the software package and relevant artifacts stored in the software artifact set that are required to perform a test of the software package binaries **142**. The testing environment for the test for the software package may be created inside the testing system **140** or may be generated in a resource outside of the vehicle software release management system **100**. In the latter scenario, the testing system may send relevant software artifacts to test the software, including any relevant test scripts, dependencies, and/or software package binaries. Once the test has been performed according to the test plan, the vehicle testing system **140** may store the resulting testing metadata in the associated software artifact set. The certification response may be added to the relevant software artifact set.

(22) In some embodiments, the vehicle software distribution system **160** of the vehicle software release management system **100** may receive a request for deployment of the vehicle software package **162**. The vehicle software distribution system may obtain the correct software artifact set associated with the indicated deployment request, verify that it has been certified, and transmit the certified vehicle software package in the required format **164** via a distribution adapter **170**. In some embodiments, the certified vehicle software package may be sent to multiple destinations, such as vehicle A **180**, vehicle B **182**, and vehicle N **184**. Although FIG. 1 depicts only vehicles as the destination of the certified vehicle software package, the software package may be requested to be deployed to any component involved in a vehicle data pipeline. For example, in some embodiments, the vehicle software package may be sent to a vehicle edge distribution location, a vehicle data pipeline, a vehicle data late, or other components involved in the vehicle data pipeline, as further discussed in FIG. 6.

(23) FIG. 2 illustrates a more detailed view of a vehicle software release management system that receives a vehicle software package and a test plan to generate a versioned artifact set for a vehicle model that may be stored in a versioned artifact set storage, according to some embodiments.

(24) In FIG. 2, the release management interface **110** may receive the vehicle software package **111** as described in FIG. 1. In some embodiments, the received vehicle software package may be a software package **200** for a given vehicle model. The software package **200** may be for one or more components having a given vehicle specification, such as a specific vehicle make and/or model. In some embodiments, the software package **200** may be for multiple vehicle models and may be deployable to multiple vehicle models. In some embodiments, the vehicle software package **200** may not be for a vehicle but for a component of a vehicle data pipeline or vehicle data stream generation process. For example, the software package may be related to software used in a vehicle data lake. The software package **200** may be software used in the operation of a range of components in vehicles as well as a range of components in data pipeline of vehicles.

(25) In some embodiments, the vehicle software release management system **100** may be configured to verify credentials **202** and determine whether the vehicle software package **200**, or any other information sent to the vehicle software release management system **100**, has proper authorization. In some embodiments, the vehicle software release management system **100** may process credentials sent with the software package **200**. For example, the software release management system **100** may verify HTTP authentication credentials such as username and password sent alongside an API call used to send the software package **200**. In some embodiments, various other authentication techniques, such as key authentication using unique keys disclosed to authorized users or OAuth frameworks may be used to verify credentials **202** of various requests. In some embodiments, the request to receive the software package **200** may result in a lookup on a username, username password, role, or other identity of the software package **200** sender. Unauthorized software packages, requests to receive software packages, or request to deploy/roll back software packages may be denied upon determination that the request has improper access credentials. The term “access credentials” or “credentials” are used herein to mean various credentials that verify the identity of the requesting entity and/or authorization granted to the requesting entity that is required to perform the requested action. For example, in some embodiments, the software management system **100** may reject a request made by a user having a verified identity (e.g., known username and password) because the entity associated with the verified identity has lacks authorization to access data targeted by the request, e.g., the entity has improper “access credentials” based on a lack of authorization to perform the request.

(26) The vehicle software release management system **100** may send a notification of the denial of permissions to vehicle software release management system **100** managers. The vehicle software release management system **100** may verify the credentials of the request to indicate a given vehicle model of the software package **113** and the request to receive a test plan **114** in a manner similar to the techniques described above. In some embodiments, the given vehicle model for the software package may be indicated in the software package **200** itself or in the request sent to the vehicle software release managements system **100** to receive the software package **200**.

(27) In some embodiments, the simulation configuration interface **120** may receive a test plan **210** to run tests for a given software package for a given vehicle model. The test plan **210** may be associated with the software package **200** and may describe the test strategy, objectives, and resources required in testing the vehicle software package **200**. Various testing information may be provided to the simulation configuration interface **120** through the test plan **210**. Testing information provided through the test plan **210** may include configuration files of the testing environment, configuration files of the software package **200**, resource files for the testing environment, resource files for the software package **200**, and/or a test script **250**. In some embodiments, the simulation configuration interface **120** may use a test script generator **212** to generate a test script **250** according to the test plan. In some embodiments, the test configuration files, and the test resources may be generated by the simulation configuration interface **120** according to the received test plan **210**. The test script **250** may be one or more pieces of software that are to be run during the testing of the software package **200**.

(28) Various components of the software artifact set may be generated by the release management interface **110**, versioned artifact set generation module **112**, and/or the simulation configuration interface **120**. For example, test script generator **212** may generate a test script according to the received test plan **210**. The versioned artifact set generation module **112** may obtain the various software artifacts, including the software package **200** and the test plan **210** to generate a software artifact set **125**. In some embodiments, a software artifact set for a vehicle model **230** may be generated by the versioned artifact set generation module **112** and stored in the versioned artifact set storage **130**. The software artifact set for the vehicle model **230** may include the received software package **200**, test plan **242**, test configurations **246**, test scripts **250**, vehicle software binaries **240**, software dependencies **244**, and software configuration files **248**. The test configurations **246** may define parameters, options, settings and/or other preferences of tests to be run on the software package, and may furthermore describe the testing environment that the software packages are to be tested on. In some embodiments, the test configurations **246** may be generated by the versioned artifact set generation module **112** in accordance with the test plan **242**. In some embodiments, the vehicle software binaries **240** may be generated by the versioned artifact set generation module **112**. The software binaries **240** may be compiled code that allows the software package **200** to be installed without having to compile the source code in the destination. The software binaries **240** may be in a format that allows a vehicle destination to apply the code without a separate compilation step. In some embodiments, the software dependencies **244** may be various code libraries or packages, that the vehicle software package **200** depends upon and that are reused in the execution of the vehicle software package **200**. The software dependencies **244** may be direct dependencies that the software package **200** calls directly in its code, or they may be transitive dependencies that a dependency of the software package **200** is dependent upon. The versioned artifact set generation module may be provided one or more portions of the software dependencies **244** with the software package **200** in the initial request to receive the software package, or the versioned artifact set generation module **112** may determine the software dependencies **244** based on analysis of the received software package **200**.

(29) FIG. 3 illustrates a more detailed view of a vehicle software release management system that initiates a workflow to perform testing of a vehicle software package according to the associated software artifact set for a vehicle model and storing the vehicle software test metadata in the software artifact set for the vehicle model, according to some embodiments.

(30) In some embodiments, the receipt of a vehicle software package **111** as described in FIG. 1 may trigger the release management interface **110** to initiate a workflow **310** which may include performing one or more operations to test the received vehicle software package **200**. A workflow engine **320** may initiate the testing of the vehicle software package **325** using the testing system **140**. The testing system **140** of the vehicle software release management system **100** may obtain **220** information required to test the vehicle software package **330** in the software artifact set for a vehicle model **230**—the required testing information may include the software package **200**, the vehicle software binaries **240**, the test plan **242**, the test configurations **246**, and/or the test script **250**.

(31) In some embodiments, testing environment **332** may be implemented by the provider network of the vehicle software release management system **100**. Testing environment **332** may offer instances, containers, and/or functions according to various configurations for testing including the performance of the test script **250**. In some embodiments, testing environment **332** may, for example, comprise one or more servers with a specified computational capacity (which may be specified by indicating the type and number of CPUs, the main memory size, and so on, for example to replicate conditions of an electronic control unit (ECU) of a vehicle) and a specified software stack (e.g., a particular version of an operating system, as may be used in a vehicle environment, such as on an ECU). A testing environment **332** may provide a virtual operation system or other operating environment for executing or implementing the vehicle software package

200. The vehicle software package **200** may be implemented as one or more operations that are performed upon request or in response to an event, and the testing environment **332** may be automatically scaled to provide the appropriate number and/or type of computing resources to perform the operations in accordance with the testing requirements described in the test plan **242**. A number of different types of computing devices may be used singly or in combination to implement the testing environment **332**, including general purpose or special purpose computer servers, storage devices, network devices and the like. In some embodiments, the testing environment **332** may be implemented as part of the testing system **140** of the vehicle software release management system **100**. In some embodiments, the computing devices used to implement the testing environment **332** may be cloud resources that are configured to simulate vehicle resources, such as ECUs, etc.

(32) Various compute instances, containers, and/or functions may be used to operate or implement a variety of different testing environments, such as operating systems of various vehicle models or components. The testing environment **332** may support various interpreted or compiled programming languages such as Ruby, Perl, Python, C, C++ and the like, or high-performance computing services suitable for performing the software package testing. In some embodiments, testing environment **332** may be configured according to test configurations **246**. For example, the size of compute instances, containers, and/or functions, such as a particular number of virtual CPU cores, memory, cache, storage, as well as any other performance characteristic may be determined by the test configurations **246** (for example in order to simulate vehicle systems). In some embodiments, the test configurations **246** may determine properties of the electronic control units simulated in the testing environment **332** of the vehicle model that the software package **200** is intended to be deployed to. For example, the test configuration **246** may determine the operating system of the electronic control unit being simulated in the testing environment **332**.

Configurations of the compute instances, containers, and/or functions of the testing environment **332** may furthermore be determined by a request to configure the testing system **350**. In some embodiments, the simulation configuration interface **120** may receive a request to configure the testing system **350**, and upon verification of correct credentials **202**, may configure the testing system **354** according to the received request. In various embodiments, the software package **200** and/or vehicle software binaries **240** may be implemented by a particular node or may be distributed across several nodes. In some embodiments, a given node may implement the functionality of more than one service system component (e.g., more than one data store component).

(33) Once the testing system **140** has performed the test of the software package/software package binaries **142**, the testing system **140** may receive a testing result **334** documenting the results of the test according to the test plan. The test results may comprise one or more statistics measuring the performance of the vehicle software package **200** and/or determinations whether the software package **200** meets the requirements of the software testing described in the test plan **242**. In some embodiments, the results of the test script **250** used to test the software package **200** may also be returned to the testing system **140**. Once the test result has been received by the testing system **140**, in some embodiments, the testing system **140** may store **336** vehicle software test metadata containing the received test results in the software artifact set for the vehicle model **230**. The software artifact set **230** may store the test results in the vehicle software test metadata **340**. In some embodiments, multiple iterations of the test having different test plans may be performed upon a request to configure testing system **350** and the results of the respective runs may be stored in the software artifact set **230** as part of the vehicle software test metadata **340**. In some embodiments, a particular iteration of the tests may generate a different software artifact set for the vehicle model associated with a particular software test iteration.

(34) FIG. 4 illustrates a more detailed view of a vehicle software release management system that initiates a workflow to perform certification of a vehicle software package according to a

corresponding software artifact set for a particular vehicle model. The vehicle software release management system also stores the vehicle software certification metadata in the software artifact set for the vehicle model, according to some embodiments.

(35) In some embodiments, as part of the workflow initiated by the release management interface **110** as discussed in FIG. **1**, the workflow engine **320** may initiate certification of the vehicle software package **400**. In some embodiments, the certification system **150** may obtain information to certify the software package **412** from the software artifact set associated with the software package **200**. The information required to certify the vehicle software package may include results of the vehicle software testing obtained from the vehicle software test metadata **340**, such as pass/fail results of test script **250** and/or performance statistics of key metrics obtained from the test plan. In some embodiments, the certification system **150** may send the certification request **414** using a certification adapter **416** to translate the request into a format acceptable to a certification destination **418**. In some embodiments, the certification adapter **416** may provide programmatic interfaces (e.g., APIs, web pages or web sites, graphical user interfaces, or command-line tools) to format and send the certification request **414**.

(36) In some embodiments, the certification destination **418** may send a certification response to the certification system **150**. The certification system **150** may receive certification response **420** from the certification destination **418** that indicates whether the software package is approved to be deployed to the given vehicle model. The certification system **150** may receive a rejection of certification as its certification response along with, in some embodiments, information regarding the reason for the rejection of certification. The certification system **150** may store a certification record **422** with the certification response in the software artifact set **230** upon receipt of the certification response. In some embodiments, the certification destination **418** may be used to initiate a certification workflow separate from the workflow in the vehicle software release management system **100** in order to generate the certification response. In some embodiments, the certification record **430** stored in the software artifacts set **230** may be used to determine which one of the software packages stored in the versioned artifact set storage **130** may be deployed to a software package destination.

(37) FIG. **5** illustrates a more detailed view of a vehicle software release management system that initiates a workflow to deploy a vehicle software package selected from a versioned artifact set storage to a software package destination, according to some embodiments.

(38) In some embodiments, the release management interface **110** may receive a request to deploy a selected one of vehicle software package **502**. The request may indicate a software package to deploy from the one or more software artifact set(s) **230** stored in the versioned artifact set storage **130**. The release management interface **110** may, in some embodiments, deploy the software package as part of the workflow initiated by the release management interface **110** upon receiving a vehicle software package, as discussed in FIG. **1**. The workflow engine **320** may initiate deployment of the vehicle software package **500** and instruct the distribution system **160** to obtain information to deploy the software package **510** from the software artifact set associated with the software package **200** selected in a request to deploy. In some embodiments, the information required to deploy the software package may include the software package **200** itself and/or the vehicle software binaries **240**. In some embodiments, the distribution system **150** may transmit a certified vehicle software package in the required format **164** to one or more software package destination(s) **520** using a distribution adapter **416** that translates the request into a format acceptable to the software package destination(s) **418**. In some embodiments, the distribution adapter **416** may provide programmatic interfaces (e.g., APIs, web pages or web sites, graphical user interfaces, or command-line tools) to format and transmit the certified vehicle software package.

(39) In some embodiments, the software package destination **520** may be a vehicle, vehicle component, or components involved in the vehicle data pipeline. The certified vehicle software

package destination selected for the vehicle software package to be transmitted to may include one or more vehicles, a vehicle edge distribution **522**, a vehicle data pipeline **524**, a vehicle data lake **526**, etc. In some embodiments, the vehicle edge distribution **522** may further format the transmitted vehicle software package and handle the distribution to various vehicles instead of the distribution system **150** or the distribution adapter **170**. The selected vehicle software package **200** and/or the vehicle software binaries **240** may be sent to multiple software package destinations **520**. For example, the software package **200** may be sent to multiple vehicles, multiple vehicle data pipelines, multiple data lakes etc.

(40) FIG. **6** illustrates a more detailed view of a vehicle software release management system that receives a request to rollback a vehicle software package and initiates a workflow to transmit a vehicle software package from a software artifact set that was previously deployed or roll back software resources from a software package destination (such as one or more previously deploy vehicle software packages), according to some embodiments.

(41) In some embodiments, the release management interface **110** may receive a request to roll back a deployed vehicle software package **600**. The request to rollback the vehicle software package **600** may trigger a workflow **602** and cause the workflow engine **320** to initiate rollback of the vehicle software **604** that has been deployed to one or more software package destinations **520** as described in FIG. **5**. The vehicle software release management system, may in some embodiments, transmit instructions to remove the portions attributed to the software package requested to be rolled back instead of providing a replacement software package. In some embodiments, the rollback request may include a selection from one of the deployed software artifact sets from which a replacement software package may be transmitted to one or more software package destination(s) **520**. For example, the versioned artifact set storage **130** may store software artifact sets of deployed software packages **610** that contain multiple previously deployed software artifact sets. The software artifact sets of deployed software packages **610** may include a deployed software artifact set A **612**, a deployed software artifact set B **616**, and a deployed software artifact set N **620**, each having a different version of the software package.

(42) In some embodiments, the rollback request may be a request to roll back resources/portions of the deployed software package to revert back to a previous software package version. In embodiments that require transmitting a replacement software package, the distribution system may obtain information to deploy a selected vehicle software package **624**, such as obtaining the software package or vehicle software binaries of the selected software artifact set. In some embodiments, the distribution system **160** may determine which one of the deployed software artifact sets is to be selected for roll back based on the respective certification records. For example, software package A **614**, certification record for software package B **618**, or certification record for software package N **622** of the respective deployed software artifact sets may be analyzed to determine which one of the software packages is to be transmitted. As discussed in FIG. **5**, in some embodiments, the software package destination(s) **520** may be any vehicle, vehicle component, or components involved in the vehicle data pipeline. The replacement vehicle software package may be transmitted **626** or the deployed vehicle software package may be rolled back from a vehicle, vehicle edge distribution **522**, vehicle data pipeline **524**, and/or vehicle data lake **526**. In some embodiments, the distribution adapter **170** may further format the replacement vehicle software package to roll back to or handle the roll back of various vehicles instead of the distribution system **150**.

(43) FIG. **7** illustrates a flowchart of operations performed by a vehicle software release management system to receive a vehicle software package and a test plan, generate a versioned artifact set for the vehicle software package, initiate performance of testing, and send request to certify the software package according to the software artifact set, according to some embodiments.

(44) At block **710** a vehicle software release management system of a vehicle software release management system receives a vehicle software package. As discussed in FIG. **2**, the software

package may be for one or more components of a specific vehicle specification such as a certain make and/or model. In some embodiments, the vehicle software package may not be for a vehicle but for a component of a vehicle data pipeline having a certain model (e.g., the software package may be related to software used in a vehicle data lake).

(45) At block **720**, the vehicle software release management system receives a test plan for the vehicle software package. The test plan may be associated with the software package received at block **710** and may describe the test strategy, objectives, and the resources required in testing the vehicle software package. As discussed in FIG. 2, various other testing information may be generated using the test plan, including configuration files of the testing environment and/or the software package, resource files for the testing environment and/or the software package, and test scripts, in some embodiments.

(46) At block **730**, the vehicle software release management system receives, or determines, a given vehicle model of one or more vehicle models to which the vehicle software package is to be deployed. In some embodiments, the given vehicle model for the software package may be indicated in the software package itself or in the request by which the software package is received. In some embodiments, the vehicle software release management system may receive the vehicle model the software package is associated with, such as from a customer, an entity submitting the software package, a database of vehicle models, etc.

(47) At block **740**, the vehicle software release management system generates a software artifact set for the vehicle software package having vehicle software binaries for the given vehicle model and instructions to configure test resources for the given vehicle model, wherein the instructions are generated based on the test plan. In some embodiments, the vehicle software binaries may be generated by a versioned artifact set generation module of the vehicle release software management service. In some embodiments, the versioned artifact set generation module may further obtain software dependencies that the package depends upon or generate other software artifacts to include in the set, as discussed in FIG. 2.

(48) At block **750**, the vehicle software release management system stores the generated software artifact set. In some embodiments, the software artifact set may be stored in a versioned artifact set storage. As discussed in FIG. 1, the versioned artifact set storage may contain multiple software artifact sets for each one of the received vehicle software packages.

(49) At block **760**, the vehicle software release management system initiates performance of a vehicle software test of the vehicle software binaries of the software artifact set based on the test plan. In some embodiments, a testing system of the vehicle software release management system may obtain information to test the vehicle software package from the software artifact set for a vehicle model, as discussed in FIG. 3. In some embodiments, the testing system may obtain the software package and/or the vehicle software binaries to test in a testing environment as described by the test plan or test configurations of the software artifact set.

(50) At block **770**, the vehicle software release management system sends a request to certify the vehicle software package having the software artifact set and metadata generated by the vehicle software test for the given vehicle model. The information required to certify the vehicle software package may include results of the vehicle software testing obtained from the vehicle software test metadata such as pass/fail results of test script, performance statistics of key metrics obtained from the test plan, etc. as further described in FIG. 4.

(51) FIG. 8 illustrates a flowchart of operations performed by a vehicle software release management system to initiate deployment of a certified vehicle software package to a vehicle software package destination, according to some embodiments.

(52) At block **810** a vehicle software release management system stores the software artifact set in a software artifact set storage having a plurality of certified vehicle software artifact sets associated with a plurality of certified vehicle software packages for the given vehicle model.

(53) At block **820** a vehicle software release management system selects a certified software

package of the plurality of certified vehicle software packages in a required format to be deployed by a distribution adapter. In some embodiments, a distribution adapter of the vehicle software release management system may translate the request into a format acceptable to the software package destination. In some embodiments, the distribution adapter may provide programmatic interfaces (e.g., APIs, web pages or web sites, graphical user interfaces, or command-line tools) to format and transmit the certified vehicle software package as described in FIG. 5.

(54) At block **830**, the vehicle software release management system transmits the selected certified software package that has been prepared in the required format to the vehicle software package destination. In some embodiments, the software package destination may be any one of a vehicle, a vehicle component, or components involved in the vehicle data pipeline. For example, as discussed in FIG. 5, the certified vehicle software package selected that is transmitted may be one or more vehicles, a vehicle edge distribution, a vehicle data pipeline, or a vehicle data lake. In some embodiments, the selected certified software package may be sent to multiple locations including multiple vehicles, multiple vehicle data pipelines, multiple data lakes etc.

(55) FIG. 9 illustrates a flowchart of operations performed by a vehicle software release management system to initiate rollback of a deployed vehicle software package from a vehicle software package destination, according to some embodiments.

(56) At block **910** a vehicle software release management system receives a request to roll back a software package deployed to the software package destination to a replacement certified software package selected from a set of software packages stored in the certified software artifact set storage. In some embodiments, the rollback request may include a selection from one of the deployed software artifact sets from which the replacement software package may be transmitted to one or more software package destination to be replaced.

(57) At block **920**, the vehicle software release management system determines the replacement certified software package to be deployed as part of performing the roll back from the set based on certification records stored in the certified software artifact set storage.

(58) At block **930**, the vehicle software release management system sends the replacement certified vehicle software package to the vehicle software package destination, wherein the deployment adapter prepares the replacement certified software in the required format required by the vehicle software package destination. As discussed in FIG. 5, in some embodiments, the software package destination may be any vehicle, vehicle component, or components involved in the vehicle data pipeline. For example, the replacement vehicle software package may be to vehicle edge distribution, vehicle data pipeline, and/or vehicle data lake.

Example Computer System

(59) FIG. 10 illustrates exemplary computer system **1000** usable to implement the vehicle software release management system as described above with reference to FIGS. 1-9. In different embodiments, computer system **1000** may be any of various types of devices, including, but not limited to, a network computer, a mobile device, a consumer device, application server, storage device, a peripheral device such as a switch, modem, router, or in general any type of computing or electronic device.

(60) Various embodiments of program instructions for receiving a vehicle software package, generating a software artifact set, initiating workflows to test and certify the vehicle software package, as described herein, may be executed in one or more computer systems **1000**, which may interact with various other devices. Note that any component, action, or functionality described above with respect to FIGS. 1-9 may be implemented on one or more computers configured as computer system **1000** of FIG. 10, according to various embodiments. In the illustrated embodiment, computer system **1000** includes one or more processors **1010** coupled to a system memory **1020** via an input/output (I/O) interface **1030**. Computer system **1000** further includes a network interface **1040** coupled to I/O interface **1030**, and one or more input/output devices **1050**, such as cursor control device **1060**, keyboard **1070**, and display(s) **1080**. In some cases, it is

contemplated that embodiments may be implemented using a single instance of computer system **1000**, while in other embodiments multiple such computer systems, or multiple nodes making up computer system **1000**, may be configured to host different portions or instances program instructions as described above for various embodiments. For example, in one embodiment some elements of the program instructions may be implemented via one or more nodes of computer system **1000** that are distinct from those nodes implementing other elements.

(61) In some embodiments, computer system **1000** may be implemented as a system on a chip (SoC). For example, in some embodiments, processors **1010**, memory **1020**, I/O interface **1030** (e.g., a fabric), etc. may be implemented in a single SoC comprising multiple components integrated into a single chip. For example, an SoC may include multiple CPU cores, a multi-core GPU, a multi-core neural engine, cache, one or more memories, etc. integrated into a single chip. In some embodiments, an SoC embodiment may implement a reduced instruction set computing (RISC) architecture, or any other suitable architecture.

(62) System memory **1020** may be configured to store compression or decompression program instructions **1022** and/or sensor data accessible by processor **1010**. In various embodiments, system memory **1020** may be implemented using any suitable memory technology, such as static random-access memory (SRAM), synchronous dynamic RAM (SDRAM), nonvolatile/Flash-type memory, or any other type of memory. In the illustrated embodiment, program instructions **1022** may be configured to implement any of the functionality described above. In some embodiments, program instructions and/or data may be received, sent or stored upon different types of computer-accessible media or on similar media separate from system memory **1020** or computer system **1000**.

(63) In one embodiment, I/O interface **1030** may be configured to coordinate I/O traffic between processor **1010**, system memory **1020**, and any peripheral devices in the device, including network interface **1040** or other peripheral interfaces, such as input/output devices **1050**. In some embodiments, I/O interface **1030** may perform any necessary protocol, timing or other data transformations to convert data signals from one component (e.g., system memory **1020**) into a format suitable for use by another component (e.g., processor **1010**). In some embodiments, I/O interface **1030** may include support for devices attached through various types of peripheral buses, such as a variant of the Peripheral Component Interconnect (PCI) bus standard or the Universal Serial Bus (USB) standard, for example. In some embodiments, the function of I/O interface **1030** may be split into two or more separate components, such as a north bridge and a south bridge, for example. Also, in some embodiments, some or all of the functionality of I/O interface **1030**, such as an interface to system memory **1020**, may be incorporated directly into processor **1010**.

(64) Network interface **1040** may be configured to allow data to be exchanged between computer system **1000** and other devices attached to a network **1085** (e.g., carrier or agent devices) or between nodes of computer system **1000**. Network **1085** may in various embodiments include one or more networks including but not limited to Local Area Networks (LANs) (e.g., an Ethernet or corporate network), Wide Area Networks (WANs) (e.g., the Internet), wireless data networks, some other electronic data network, or some combination thereof. In various embodiments, network interface **1040** may support communication via wired or wireless general data networks, such as any suitable type of Ethernet network, for example; via telecommunications/telephony networks such as analog voice networks or digital fiber communications networks; via storage area networks such as Fiber Channel SANs, or via any other suitable type of network and/or protocol.

(65) Input/output devices **1050** may, in some embodiments, include one or more display terminals, keyboards, keypads, touchpads, scanning devices, voice or optical recognition devices, or any other devices suitable for entering or accessing data by one or more computer systems **1000**. Multiple input/output devices **1050** may be present in computer system **1000** or may be distributed on various nodes of computer system **1000**. In some embodiments, similar input/output devices may be separate from computer system **1000** and may interact with one or more nodes of computer system **1000** through a wired or wireless connection, such as over network interface **1040**.

(66) As shown in FIG. 10, memory 1020 may include data storage 1035 and may include program instructions 1025, which may be processor-executable to implement any element or action described above. In one embodiment, the program instructions may implement the methods described above. In other embodiments, different elements and data may be included.

(67) Computer system 1000 may also be connected to other devices that are not illustrated, or instead may operate as a stand-alone system. In addition, the functionality provided by the illustrated components may in some embodiments, be combined in fewer components or distributed in additional components. Similarly, in some embodiments, the functionality of some of the illustrated components may not be provided and/or other additional functionality may be available.

(68) Those skilled in the art will also appreciate that, while various items are illustrated as being stored in memory or on storage while being used, these items or portions of them may be transferred between memory and other storage devices for purposes of memory management and data integrity. Alternatively, in other embodiments some or all of the software components may execute in memory on another device and communicate with the illustrated computer system via inter-computer communication. Some or all of the system components or data structures may also be stored (e.g., as instructions or structured data) on a computer-accessible medium or a portable article to be read by an appropriate drive, various examples of which are described above. In some embodiments, instructions stored on a computer-accessible medium separate from computer system 1000 may be transmitted to computer system 1000 via transmission media or signals such as electrical, electromagnetic, or digital signals, conveyed via a communication medium such as a network and/or a wireless link. Various embodiments may further include receiving, sending or storing instructions and/or data implemented in accordance with the foregoing description upon a computer-accessible medium. Generally speaking, a computer-accessible medium may include a non-transitory, computer-readable storage medium or memory medium such as magnetic or optical media, e.g., disk or DVD/CD-ROM, volatile or non-volatile media such as RAM (e.g., SDRAM, DDR, RDRAM, SRAM, etc.), ROM, etc. In some embodiments, a computer-accessible medium may include transmission media or signals such as electrical, electromagnetic, or digital signals, conveyed via a communication medium such as network and/or a wireless link.

(69) The methods described herein may be implemented in software, hardware, or a combination thereof, in different embodiments. In addition, the order of the blocks of the methods may be changed, and various elements may be added, reordered, combined, omitted, modified, etc. Various modifications and changes may be made as would be obvious to a person skilled in the art having the benefit of this disclosure. The various embodiments described herein are meant to be illustrative and not limiting. Many variations, modifications, additions, and improvements are possible. Accordingly, plural instances may be provided for components described herein as a single instance. Boundaries between various components, operations and data stores are somewhat arbitrary, and particular operations are illustrated in the context of specific illustrative configurations. Other allocations of functionality are envisioned and may fall within the scope of claims that follow. Finally, structures and functionality presented as discrete components in the example configurations may be implemented as a combined structure or component. These and other variations, modifications, additions, and improvements may fall within the scope of embodiments as defined in the claims that follow.

Claims

1. A method, comprising: receiving a vehicle software package to be deployed to one or more vehicle models; receiving a test plan for the vehicle software package; generating, using a vehicle software release management system, a software artifact set for the vehicle software package, wherein the vehicle software release management system is implemented using one or more

computing devices comprising one or more processors and memory, wherein the software artifact set comprises vehicle software binaries for a given one of the one or more vehicle models and test resources for a given vehicle model and instructions to configure test resources for the given vehicle model, wherein the instructions are generated based on the test plan, and wherein the test resources are generated based on the test plan; performing a vehicle software test of the vehicle software binaries of the software artifact set based on a test plan for the given vehicle model; sending a request to certify the vehicle software package for the given vehicle model, wherein the request comprises the software artifact set and metadata generated by the vehicle software test of the vehicle software binaries for the given vehicle model; and storing, upon certification of the vehicle software package, the software artifact set in a certified software artifact set storage, wherein the stored software artifact set comprises the vehicle software binaries for the given vehicle model and the instructions to configure test resources for the given vehicle model used in the performed vehicle software test.

2. The method of claim 1, further comprising: storing, upon certification of the vehicle software package, the software artifact set and an associated certification record in the certified software artifact set storage configured to store software artifact sets and associated certification record for respective vehicle software packages that have been certified; selecting, or receiving a request to select, a certified vehicle software package associated with a certified software artifact set in the certified software artifact set storage to be deployed by a deployment adapter of the vehicle software release management system, wherein the one or more computing devices implement the deployment adapter, and wherein the deployment adapter prepares the selected certified software package in a required format required by a vehicle software package destination; and transmitting the selected certified vehicle software package that has been prepared in the required format to the vehicle software package destination.

3. The method of claim 2, further comprising: storing one or more additional software artifact sets in the certified software artifact set storage, wherein respective ones of the one or more additional software artifact sets further have respective associated certification records; selecting, or receiving a request to select, one or more additional certified software packages associated with one or more certified software artifact sets and certification records stored in the certified software artifact set storage to be deployed by the deployment adapter, wherein the deployment adapter prepares the selected one or more additional certified software packages in the required format required by the vehicle software package destination; and transmitting the selected one or more additional certified vehicle software packages that have been prepared in the required format to the vehicle software package destination.

4. The method of claim 3, further comprising: receiving a request to roll back a software package deployed to the software package destination to a replacement certified software package, wherein the replacement software package is selected from a set of software packages stored in the certified software artifact set storage, wherein the set comprises the deployed certified vehicle software package and the one or more additional deployed certified vehicle software packages; determining the replacement certified software package to be deployed as part of performing the roll back from the set based on certification records stored in the certified software artifact set storage; and sending the replacement certified vehicle software package to the vehicle software package destination, wherein the deployment adapter prepares the replacement certified software in the required format required by the vehicle software package destination.

5. The method of claim 2, wherein the vehicle software package destination is a vehicle edge distribution system configured to distribute vehicle software packages to vehicles of the given vehicle model, or a vehicle data pipeline for vehicle data from vehicles of the given vehicle model, or a data lake storing vehicle data from vehicles of the given vehicle model.

6. The method of claim 2, further comprising: identifying a level of authorization associated with an entity submitting the vehicle software package; and determining whether the authorization level

is sufficient for generating a software artifact set for the vehicle software package, wherein generating the software artifact set for the vehicle software package is based on the determination that the authorization level of the entity submitting the vehicle software package is sufficient.

7. The method of claim 1, wherein the method comprises: receiving a test plan for the vehicle software package; and generating a test script according to the test plan wherein the vehicle software testing system performs the test script as part of the vehicle software test, and wherein the metadata generated by the vehicle software test includes results of the test script.

8. The method of claim 1, wherein the software artifact set further comprises one or more of: system mappings to a deployment destination, one or more configuration files, one or more resource files, or one or more test scripts.

9. The method of claim 1, wherein the test plan further comprises one or more of: one or more configuration files, an electronic control unit configuration, or one or more dependencies required to execute the vehicle software test.

10. One or more non-transitory, computer-readable storage media, storing program instructions that when executed on or across one or more processors cause the one or more processors to implement a vehicle software release management service that implements: receiving a vehicle software package to be deployed to one or more vehicle models; receiving a test plan for the vehicle software package; generating a software artifact set for the vehicle software package, wherein the software artifact set comprises vehicle software binaries for a given one of the one or more vehicle models and test resources for the given vehicle model and instructions to configure test resources for the given vehicle model, wherein the instructions are generated based on the test plan, wherein the test resources are generated based on the test plan; performing a vehicle software test of the vehicle software binaries of the software artifact set based on a test plan for the given vehicle model; sending a request to certify the vehicle software package for the given vehicle model, wherein the request comprises the software artifact set and metadata generated by the vehicle software test of the vehicle software binaries for the given vehicle model; and storing, upon certification of the vehicle software package, the software artifact set in a certified software artifact set storage, wherein the stored software artifact set comprises the vehicle software binaries for the given vehicle model and the instructions to configure test resources for the given vehicle model used in the performed vehicle software test.

11. The one or more non-transitory, computer-readable storage media of claim 10, wherein the vehicle software release management service further implements: storing one or more software artifact sets and one or more associated certification records in the certified software artifact set storage; receiving a request to select one or more certified software packages associated with one or more certified software artifact sets in the certified software artifact set storage to be deployed by a deployment adapter of the vehicle software release management service, wherein the one or more non-transitory, computer-readable storage media, further store program instructions that when executed on or across the one or more processors cause the one or more processors to implement the deployment adapter, and wherein the deployment adapter prepares the selected one or more certified software packages in a required format required by a vehicle software package destination; and transmitting the selected one or more certified vehicle software packages that have been prepared in the required format to the vehicle software package destination.

12. The one or more non-transitory, computer-readable storage media of claim 11, wherein the vehicle software release management service further implements: receiving a request to roll back a deployed software package; determining a replacement certified software package from the one of one or more certified vehicle software packages stored in the certified software artifact set storage based on certification records stored in the certified software artifact set storage; and sending the replacement certified vehicle software package to the vehicle software package destination, wherein the deployment adapter prepares the replacement certified software in the required format required by a vehicle software package destination.

13. The one or more non-transitory, computer-readable storage media of claim 11, wherein the vehicle software release management service further implements: identifying an authorization level associated with the request to select the one or more certified software packages; and determining whether the authorization level is sufficient for transmitting the selected one or more certified vehicle software packages to the vehicle software package destination, wherein transmitting the selected one or more certified vehicle software packages is based on the determination that the authorization level is sufficient.

14. The one or more non-transitory, computer-readable storage media of claim 10, wherein the vehicle software package destination is a vehicle edge distribution system configured to distribute vehicle software packages to vehicles of the given vehicle model, or a vehicle data pipeline for vehicle data from vehicles of the given vehicle model, or a data lake storing vehicle data from vehicles of the given vehicle model.

15. The one or more non-transitory, computer-readable storage media of claim 10, wherein the vehicle software release management service further implements: a simulation configuration user interface configured to: receive a test plan for the vehicle software package; generate a test script according to the test plan wherein a vehicle software testing system of the vehicle software release management service performs the test script as part of the vehicle software test; and wherein the one or more non-transitory, computer-readable storage media, further store program instructions that when executed on or across the one or more processors cause the one or more processors to implement the vehicle software testing system, and wherein the metadata generated by the vehicle software test includes results of executing the test script.

16. A system, comprising: one or more computing devices each comprising one or more processors and memory storing program instructions that, when executed on or across the one or more processors, cause the one or more processors to implement a vehicle software release management system wherein the vehicle software release management system is configured to: receive a vehicle software package; receive a test plan for the vehicle software package; receive or determine a given vehicle model of one or more vehicle models to which the vehicle software package is to be deployed; generate a software artifact set for the vehicle software package, wherein the software artifact set comprises vehicle software binaries for the given vehicle model and instructions to configure test resources for the given vehicle model, wherein the instructions are generated based on the test plan; perform, via a vehicle software testing system implemented via at least one of the one or more computing devices, a vehicle software test of the vehicle software binaries of the software artifact set based on the test plan; send a request to certify the vehicle software package, wherein the request comprises the software artifact set and metadata generated by the vehicle software test of the vehicle software binaries for the given vehicle model; and store, upon certification of the vehicle software package, the software artifact set in a software artifact set storage, wherein the stored software artifact set comprises the vehicle software binaries for the given vehicle model and the instructions to configure test resources for the given vehicle model used in the performed vehicle software test.

17. The system of claim 16, wherein the vehicle software release management system is further configured to: store, upon certification of the vehicle software package, the software artifact set in the software artifact set storage comprising a plurality of certified vehicle software artifact sets associated with respective ones of a plurality of certified vehicle software packages for the given vehicle model; select, or receive a request to select, a certified software package of the plurality of certified vehicle software packages to be deployed by a deployment adapter, wherein the deployment adapter prepares the selected certified software package in a required format required by a vehicle software package destination; and transmit, via a vehicle software distribution system, the selected certified software package that has been prepared in the required format to the vehicle software package destination.

18. The system of claim 16, wherein the vehicle software testing system is configured to generate

one or more configuration files that describe a vehicle environment corresponding to the given vehicle model that the vehicle software package is to be deployed to, wherein the one or more configuration files are included in the software artifact set.

19. The system of claim 16, wherein the vehicle release management system further comprises: a simulation configuration user interface configured to: receive a test plan for the vehicle software package; and generate a test script according to the test plan, wherein the vehicle software testing system performs the test script as part of the vehicle software test, and wherein the metadata generated by the vehicle software test includes results of the test script.

20. The system of claim 16, wherein the vehicle release management system further comprises a certification system configured to: provide a certifying entity the vehicle software artifact set and the metadata generated by the vehicle software test; and receive, from the certifying entity, an indication of certification of the vehicle software package.
